

Simple Air-Quality And Gas Monitor With Notifications On Phone And Email

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The Internet of Things (IoT) is commonly used in homes and industries for monitoring sensor values. In this project, the Blynk IoT platform is used to get notifications through smartphone and email when the air-quality sensor detects a value that exceeds the user-defined limit. It can also be used as a gas leakage monitor.

Any suitable analogue sensor like MQ-135 may be connected to the analogue pin A0 of WeMos D1 board (Board1), as shown in Fig. 1. Using Blynk app we can now monitor gas or air quality through the sensor and get notification when the sensor gives a reading of more than 150. It is that simple!

Construction and testing

Download Blynk app from Google Play store and install it in your Android phone. Registration is required for its use. After registration, click on New Project and enter a project name of your choice. Under Choose Device, select 'WeMos D1' or 'WeMos D1 Mini,' as per the device you are using. Under Connection Type, select 'Wi-Fi' and click on 'Create.' An authorisation token will be sent to your registered email ID. It will enable you to see the project space on your Android smartphone. Follow the steps below for configuration.

1. In the project space, select 'Widget Box' and select 'Gauge' under Displays. You will be directed to the gauge settings screen. Under Input, select virtual pin 'V5' (if you are selecting a different virtual pin, change it in the program as well) and change the variation from the default 1023 to 200. This will set the gauge level from 0 to 200, as shown in Fig. 2.

2. Select 'Widget Box' and select

'Eventor' under Other, 'Notification' under Notifications, and 'Email' under Notifications.

3. Click on 'Eventor' in the project space and click '+ Add New Event.' Click on 'Vpin' value and select 'V5.' Choose the condition as 'is higher than' and the type as '150.'

4. Under Choose action select 'Send notification' and enter the message you

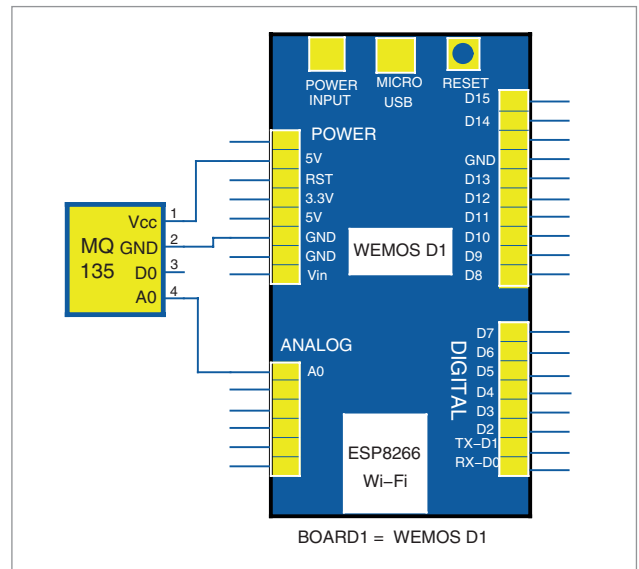


Fig. 1: Connection of MQ-135 to WeMos D1 board

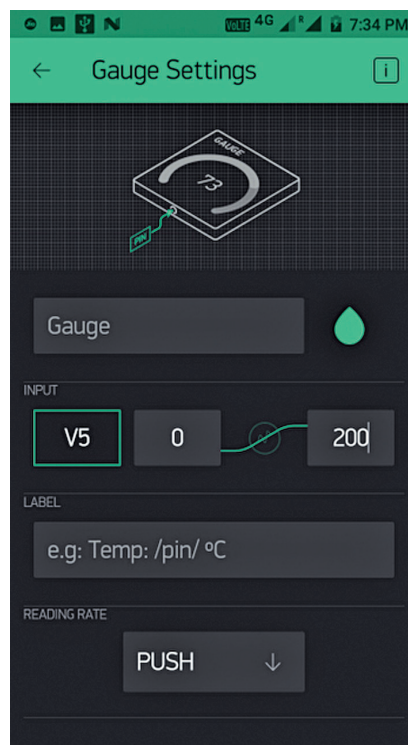


Fig. 2: Gauge settings on mobile screen

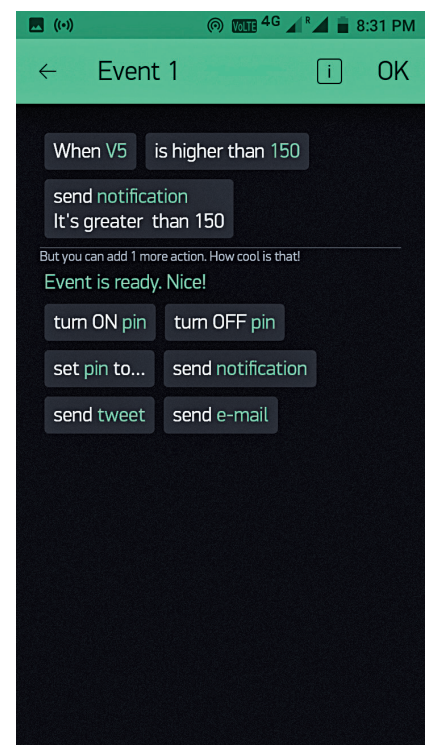


Fig. 3: Event 1 to get notification